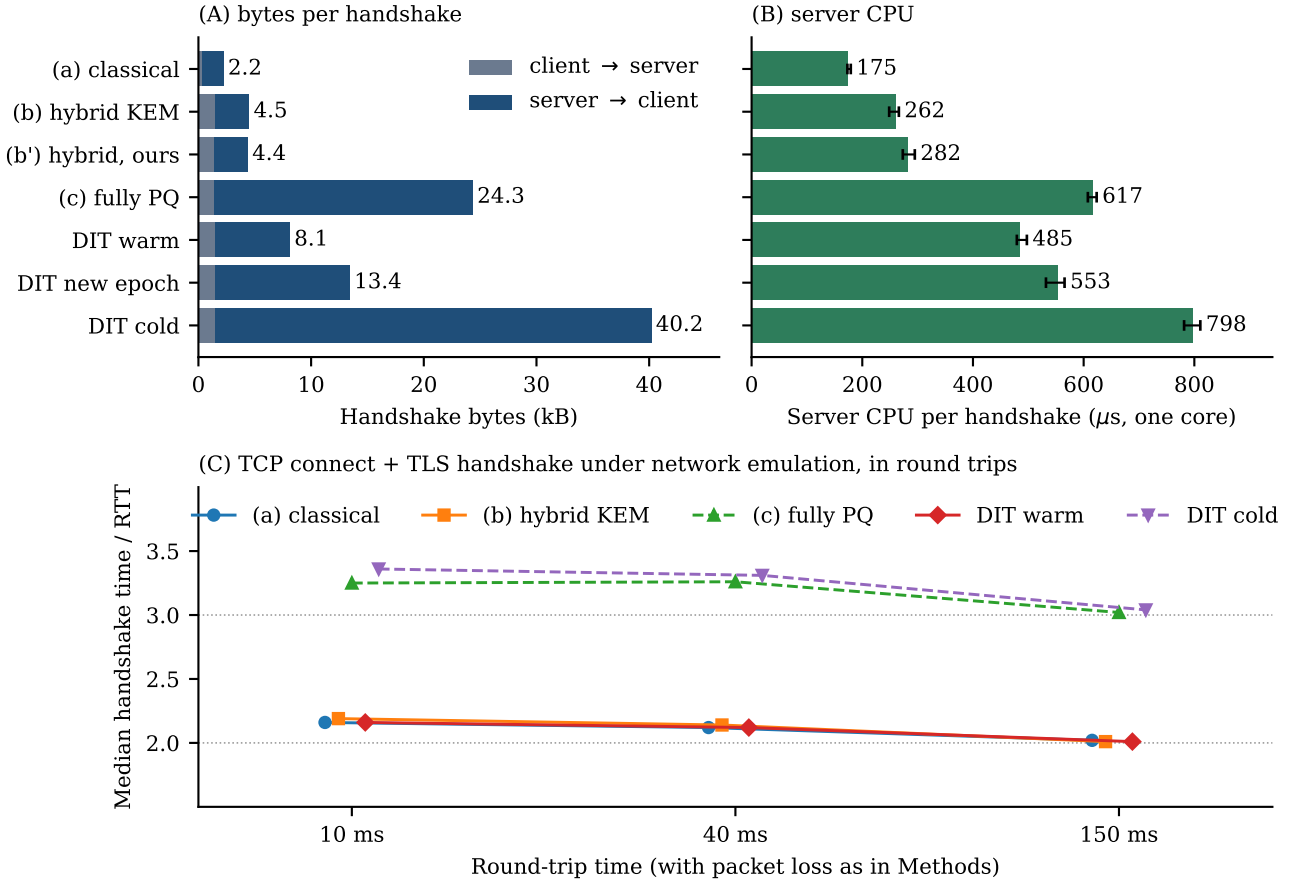
(A) bytes per handshake
(a) classical
(b) hybrid KEM
(b') hybrid, ours
(c) fully PQ
DIT warm
DIT new epoch
DIT cold
2.2
4.5
4.4
24.3
8.1
13.4
40.2
client → server
server → client
0
10
20
30
40
Handshake bytes (kB)
(B) server CPU
175
262
282
617
485
553
798
0
200
400
600
800
Server CPU per handshake (μs, one core)
(C) TCP connect + TLS handshake under network emulation, in round trips
(a) classical
(b) hybrid KEM
(c) fully PQ
DIT warm
DIT cold
Median handshake time / RTT
3.5
3.0
2.5
2.0
10 ms
40 ms
150 ms
Round-trip time (with packet loss as in Methods)